

Isabella Molina | Curriculum Vitae

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Education

Michigan State University

Ph.D in Astronomy and Astrophysics

Investigations of Shocks in Novae with Evolved Companions

East Lansing, MI

2020 - 2026

California State University, Fullerton

B.S. in Physics Cum Laude

Fullerton, CA

2017–2020

Research Experience

Michigan State University

Graduate Research Assistant

East Lansing, MI

June 2020–May 2026

Studied novae with evolved companions using radio astronomy. This includes modelling ejecta velocity, categorizing features of their blast and ejecta, and determining if they can be identified as Type Ia supernova progenitors.

IREU Albert Einstein Institute

Summer Research Experience

Hannover, Germany

May-Aug. 2018

Participated in an international research experience organized by the University of Florida. Worked with Dr. Harold Lueck, Dr. Sean Leavey, and Dr. Vaishali Adya. Built a negative impedance device to dampen motion and improve stability of mirror suspensions for laser interferometry.

California State University, Fullerton

Undergraduate Research Assistant

Fullerton, CA

August 2017–May 2020

Performed post-merger waveform analysis and parameter estimation on gravitational wave signals including binary neutron star signal GW170817. Contributed to researching the capabilities of third generation gravitational wave detectors.

Teaching Experience

University Graduate Certification in College Teaching

Michigan State University

Summer 2026

Completed Introduction to Evidence Based Undergraduate Stem Teaching online course from the Center for the Integration of Research, Teaching and Learning (CIRTL) Massive Open Online Courses (MOOCs). Currently working towards completion of a teaching portfolio.

Certificate in College Teaching Institute

Michigan State University

Summer 2025

Two day institute on completing the requirements of the MSU Certification in College Teaching

Graduate Teaching Assistant

Michigan State University

2021-2022

Taught 4 sections (totaling 320 students) of ISP 205L: Visions of the Universe, a introductory astronomy lab for non-science majors. Role included preparing and giving lectures, lab instruction, grading assignments, and holding office hours.

Supplemental Instructor

California State University, Fullerton

2018-2019

Designated tutor for undergraduate courses in classical mechanics and electricity and magnetism. Provided weekly group study sessions, prepared practice problems, attended lectures.

Rocket Class Teacher

Glendora Unified School District

April 2017

Taught a short rocketry class to 3rd and 5th graders. Students learned physics concepts behind rocketry and built and launched model rockets.

Math Success Center Tutor*Citrus Community College*

2015-2017

Tutored students on math subjects ranging from Algebra to Calculus and physics subjects including classical mechanics and electricity and magnetism.

Outreach and Mentorship

MSU Astronomy Outreach Volunteer*Michigan Recreation and Park Association*

June 2026

Volunteered for Come out & Play at the Capitol at the MSU Astronomy Outreach table. We taught kids about moon craters, what they can see in the night sky and how to use a telescope.

Undergraduate Mentor*Michigan State University*

2025 - 2026

Currently mentoring an astronomy undergraduate student at Michigan State University through an ongoing research project on classical novae with evolved companions.

Michigan State University Science Festival*Michigan State University*

April 2024

Volunteered to contribute to the preparation and implementation of an asteroid and planet formation activity.

Grad Committee for Faculty Search*Michigan State University*

2024

Interviewed prospective faculty in astronomy. Collected feedback from astronomy graduate students and delivered feedback to faculty on hiring committee.

Site Lead for the National Astronomy Consortium (NAC) Program*Michigan State University*

2023-2024

Ranked application materials of NAC applicants and helped select the NAC student for Michigan State University.

Novae Journal Club Co-Organizer*Michigan State University*

2022 - 2025

Organized time and location to host journal club. Led group discussions on selected scientific papers.

Michigan State University Public Observing Nights*Michigan State University*

2021 - Current

Volunteered as graduate student leader overseeing the public observing night, delegating tasks and organizing tours of the observatory. Also volunteered to help run the events, operating small telescopes, set up, and clean up.

Program Director PAREDS*Michigan State University*

2021 - 2025

PAREDS (Physics and Astronomy Research Experiences for Drew Scholars) is a program to increase participation in STEM by offering Michigan State University Drew Scholars paid full time summer research experiences. Paired students with physics and astronomy faculty, post docs, and graduate students. Built a supportive cohort, established mentoring relationships and provided professional development workshops and presented on career and academic topics. Organized and coordinated the PAREDS end of summer research symposium.

Awards and Certifications

Mental Health First Aid For Higher Education Certification*Michigan State University*

Spring 2026

Training to provide initial help to someone experiencing a mental health or substance use challenge. Course from the National Council for Mental Well-being.

Future Academic Scholars in Teaching (FAST) Fellowship*Michigan State University*

Fall 2025-Spring 2026

The FAST Fellowship Program gives graduate students an opportunity to have mentored teaching experiences, conduct a Teaching as Research (TAR) project and learn about teaching and assessment techniques.

Grote Reber Doctoral Fellowship Program

NRAO, Socorro, NM

Fall 2024 - Spring 2025

Pre-doctoral appointment with the National Radio Astronomy Observatory. Worked with Dr. Justin Linford to process and image Very Long Baseline Array (VLBA) radio data of the recurrent nova V3890 Sgr.

MSU Dept. of Physics and Astronomy DEI Leadership Award

Michigan State University

Spring 2024

This award recognizes excellent and innovative contributions to advancing diversity, equity, and inclusion in the Physics and Astronomy department.

Great IDEA (Inclusion, Diversity, Equity, and Accessibility) Fellowship

Michigan State University

Summer 2023

Graduate Fellowship from the College of Natural Science to support a mentored project to increase inclusion, diversity, equity and accessibility in STEM.

Packard Grant for Increasing Diversity in STEM Initiative

Michigan State University

Summer 2023

Successfully co-authored and obtained an \$85,300.00 grant for PAREDS (Physics and Astronomy Research Experiences for Drew Scholars with principle investigator Dr. Jay Strader

Michigan State University Enrichment Fellowship

Michigan State University

Fall 2020

A fellowship designated for outstanding students enrolling in a doctoral or masters program at MSU. This fellowship recognizes a students contribution to a diverse educational community, a history of overcoming challenges, high academic achievement and research goals.

Dan Black Family Fellowship Award

California State University, Fullerton

Spring 2019

Cash award given to students conducting research in the Physics Department

Louis Shapiro Award

California State University, Fullerton

Fall 2018

Cash award given to California State University Fullerton physics students who have attained upper division standing.

Dean's List

California State University, Fullerton

Fall 2017 – Spring 2020

Semester-long honor given to students who achieve a 3.5 grade point average or higher.

LSAMP Research Scholar

California State University, Fullerton

Fall 2017 - Spring 2019

Louis Stokes Alliance for Minority Participation program aims to increase STEM degrees in underrepresented students.

Technical Skills

- Observing Experience: Soar/GHTS Telescope (4.1m), Cerro Pachon, Chile
- Programming Languages / Tools: Python, Jupyter, Git, Bash/zsh, CASA, AIPS, Difmap, IRAF, LaTeX
- Major Astronomical Packages: AstroPy, Pandas, Matplotlib, Numpy, Scipy, Scikit-learn
- Other: Microsoft Office Suite, Google Workspace Tools, Stellarium, SkyNet, Zoom, Slack

Contributed Talks

Dissertation Defense

Michigan State University

May 2026

Title: "Investigations of Shocks in Novae with Evolved Companions"

Future Academic Scholars in Teaching Symposium

Michigan State University

Apr. 2026

Title: "Astronomy for non-STEM majors: Does teaching with a focus on science skills improve student learning outcomes?"

Compact Objects in Michigan and Ontario*University of Michigan-Flint**Oct. 2025*

Title: "The Recurrent Nova V3890Sgr at Radio Wavelengths"

NRAO Symposium*NRAO, Socorro, NM**Nov. 2024*

Title: "The Symbiotic Recurrent Nova V745 Sco at Radio Wavelengths"

Rise Time Conference*Purdue University**Aug. 2024*

Title: "The Symbiotic Recurrent Nova V745 Sco at Radio Wavelengths"

Radio Stars Conference*Haystack Observatory**Apr. 2024*

Title: "The Symbiotic Recurrent Nova V745 Sco at Radio Wavelengths"

Astronomy Seminar*Michigan State University**Sept. 2023*

Title: "Novae with Evolved Companions as Potential Type Ia Supernova Progenitors"

Compact Objects in Michigan and Ontario*Henry Ford College**May 2023*

Title: "Nova V745 Sco as a Potential Type Ia Supernova Progenitor"

NRAO Lunch Talk*Socorro, NM**Apr. 2023*

Title: "Nova V745 Sco as a potential Type Ia supernova progenitor"

National Society of Black Physicists*Charlottesville, VA**Nov. 2022*

Title: "Nova V745 Sco as a potential Type Ia supernova progenitor"

Poster Presentations

Great IDEAS*Michigan State University**Aug. 2023*

Title: "Physics and Astronomy Research Experiences for Drew Scholars "

American Physical Society April Meeting*Denver, CO**Apr. 2019*

Title: "Neutron Star Measurements in Third Generation Gravitational Wave Observatories"

LIGO VIRGO Collaboration Meeting*Lake Geneva, WI**Mar. 2019*

Title: "Matter Effects in Future Detectors"

American Physical Society Bridge Meeting*Stanford, CA**Dec. 2018*

Title: "Neutron Star Measurements in Third Generation Gravitational Wave Observatories"

American Physical Society Far West Section Meeting*Fullerton, CA**Oct. 2018*

Title: "Neutron Star Measurements in Third Generation Gravitational Wave Observatories"

LIGO VIRGO Collaboration Meeting*Sonoma, CA**Mar. 2018*

Title: "Matter Effects in Future Detectors"

California State University, Fullerton Physics Open House*Fullerton, CA**Aug. 2017*

Title: "Measuring Properties of Neutron Stars Using Third Generation Gravitational Wave Detectors"

California State University, Fullerton Summer Research Symposium*Fullerton, CA**Aug. 2017*

Title: "Measuring Properties of Neutron Stars Using Third Generation Gravitational Wave Detectors"

CABscon

Fullerton, CA

Nov. 2017

Title: "Measuring Properties of Neutron Stars Using Third Generation Gravitational Wave Detectors"

California State University, Fullerton Summer Research Symposium

Fullerton, CA

Aug. 2016

Title: "Measuring Properties of Neutron Stars Using Third Generation Gravitational Wave Detectors"

Publications

Peer Reviewed.....

- *Shocks in the Symbiotic Recurrent Nova V3890 Sgr: VLBI Radio Imaging and Fermi GeV Gamma-Rays*
I. Molina et al. 2026, in review
- *Near-Discovery SOAR Photometry of the Third Interstellar Object: 3I/ATLAS*
T. Frincke (including **I. Molina**) et al. 2025, 2025arXiv250902813F
- *PSR J1947-1120: A New Huntsman Millisecond Pulsar Binary*
J. Strader (including **I. Molina**) et al. 2025, Astrophysical Journal, 908, 124
- *Do Neutron Star Ultraluminous X-Ray Sources Masquerade as Intermediate-mass Black Holes in Radio and X-Ray?*
T. Panurach (including **I. Molina**) et al. 2024, Astrophysical Journal, 977, 2
- *The Symbiotic Recurrent Nova V745 Sco at Radio Wavelengths*
I. Molina et al. 2024, Monthly Notices of the Royal Astronomical Society, 534 1227P
- *Revisiting the classics: on the evolutionary origin of the 'Fe II' and 'He/N' spectral classes of novae*
E. Aydi et al. (including **I. Molina**), 2024, Monthly Notices of the Royal Astronomical Society, 527 9303P
- *Catching a nova X-ray/UV flash in the visible? Early spectroscopy of the very slow Nova Velorum 2022 (Gaia22alz)*
E. Aydi et al. (including **I. Molina**), 2024 Monthly Notices of the Royal Astronomical Society, 524 1946P
- *TESS photometry of the nova eruption in V606 Vul: asymmetric photosphere and multiple ejections?*
K. Sokolovsky et al. (including **I. Molina**), 2023, 2023arXiv231104903S
- *Multiwavelength Observations of a New Redback Millisecond Pulsar 4FGL J1910.7-5320*
K. Au et al. (including **I. Molina**), 2023, Astrophysical Journal 983, 112
- *GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral*
Physical Review Letter 119, 161101 (2017)
LIGO Scientific Collaboration and Virgo Collaboration

Non Refereed.....

- *Early Swift-XRT detection of heavily-absorbed shock-powered X-rays from nova V1935 Cen = PNV J14372177-5847400*
K. Sokolovsky et al. (including **I. Molina**), 2025, The Astronomer's Telegram, No. 17418
- *NuSTAR upper limit on X-ray emission of nova V462 Lupi*
K. Sokolovsky et al. (including **I. Molina**), 2025, The Astronomer's Telegram, No. 17281
- *VLA Proposal: Triggered Monitoring and Imaging of a New Bright Nova*
I. Molina et al. 2024, VLA/25A-409
21 hours of VLA time, Priority A
I. Molina et al. 2024, VLBA/25A-412
28 hours of VLBA time, Priority A
- *VLA Proposal: Triggered Monitoring and Imaging of a New Bright Nova*
I. Molina et al. 2023, VLA/24A-360
21 hours of VLA time, Priority A
- *VLBA Proposal: Triggered Monitoring and Imaging of a New Bright Nova*
I. Molina et al. 2023, VLBA/24A-359
21 hours of VLBA time, Priority B
- *VLA radio detection of Nova Scorpii 2024 (V1723 Sco)*

- I. Molina** et al. 2024, The Astronomer's Telegram, No. 16492
- *A Survey for Radio Emission from White Dwarfs in the VLA Sky Survey*
I. Pelisoli et al. (including **I. Molina**), 2024, ADS 2024arXiv240211015P
- *Swift X-ray detection of Nova Scorpii 2024 (V1723 Sco)*
K. Sokolovsky et al. (including **I. Molina**), 2024, The Astronomer's Telegram, No. 16484
- *NuSTAR and Swift upper limits on the early X-ray emission from Nova Scorpii 2024 (V1723 Sco, PNV J17261813-3809354)*
K. Sokolovsky et al. (including **I. Molina**), 2024, The Astronomer's Telegram, No. 16444
- *SOAR spectroscopic classification of PNV J17261813-3809354 as a nova near visible peak*
E. Aydi et al. (including **I. Molina**), 2024, The Astronomer's Telegram, No. 16440
- *SOAR spectroscopic confirmation of Gaia22bhs as a highly reddened Galactic classical nova* A. Kawash et al. (including **I. Molina**), 2022, The Astronomers Telegram, No. 15318